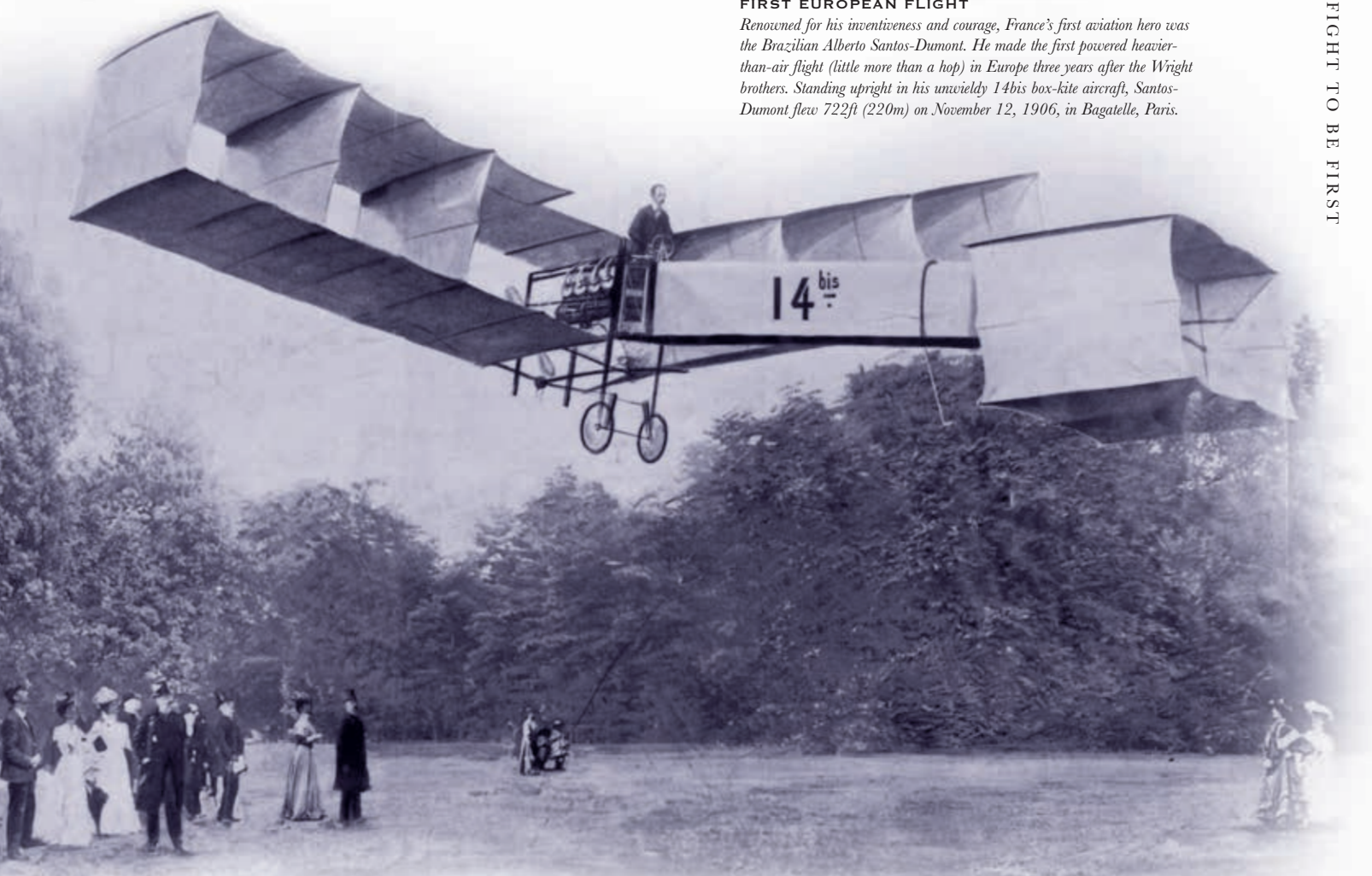


#### FIRST EUROPEAN FLIGHT

*Renowned for his inventiveness and courage, France's first aviation hero was the Brazilian Alberto Santos-Dumont. He made the first powered heavier-than-air flight (little more than a hop) in Europe three years after the Wright brothers. Standing upright in his unwieldy 14bis box-kite aircraft, Santos-Dumont flew 722ft (220m) on November 12, 1906, in Bagatelle, Paris.*



enthusiasm than method. Despite the existence of clear published accounts of the Wrights' wing-warping system, no French aviator understood the need for control in roll. Yet, stimulated by the offer of large cash prizes for variously defined "first flights" from rich enthusiasts, the French began to create successful flying machines.

The first man to claim some of this prize money was the popular Brazilian-Parisian, Alberto Santos-Dumont, already famous for his airship exploits. In 1906, Santos-Dumont built the 14bis, an ungainly, impractical biplane, with its fuselage and front elevator sticking out in front of the pilot, who stood upright in a wicker balloon basket. Its design owed much to the box kite developed by Australian Lawrence Hargrave in the 1890s – an influence that was present in many early European flying machines. Santos-Dumont's public demonstrations during the autumn of 1906 caused a sensation. Progressing from tiny hops in September to a longer hop of

about 70ft (50m) in October, he ended with a triumphant 722ft (220m) flight on November 12. Although negligible compared with the Wright brothers' flights of the previous year, Santos-Dumont's efforts were greeted in Europe as a major breakthrough. *Le Figaro* trumpeted: "What a triumph! ... The air is truly conquered. Santos has flown. Everybody will fly."

#### French engineering

French aviators had at their disposal the excellent aero-engine, the Antoinette, developed by Léon Levavasseur, and the world's first factory dedicated to aircraft

manufacture, set up by the Voisin brothers in 1906. During 1907, both Louis Blériot and Robert Esnault-Pelterie achieved short flights in tractor (powered from the front) monoplanes, a configuration that would soon play a crucial role in the evolution of flight. But the outstanding French performances of 1908 were achieved in modified Voisin biplanes. These basically

#### A FAMILY BUSINESS

*Brothers Gabriel (left) and Charles Voisin established one of the world's first airplane factories, in the Parisian suburb of Billancourt, in 1906. By 1918, it had produced over 10,000 aircraft.*

